

Ahad Khan – Computational Chemistry Projects

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Three self-directed projects forming a multi-scale study of five food antioxidants (quercetin, catechin, ascorbic acid, caffeic acid, rutin): gas-phase electronic structure (Project 1), an explicit-solvent molecular dynamics test system on ethanol–water (Project 2), and implicit continuum solvation of the same five molecules (Project 3).

Project 1: Gas-Phase DFT

OBJECTIVE

Compare gas-phase electronic reactivity descriptors across the five antioxidants and identify which is most reactive, and why.

METHODS

DFT in ORCA 6.1.1 at B3LYP/6-31G*; HOMO–LUMO gaps and conceptual-DFT descriptors; orbital and electrostatic-potential surfaces visualised in Multiwfn and ChimeraX.

RESULTS

Quercetin showed the smallest HOMO–LUMO gap (3.97 eV) of the five, consistent with its reported antioxidant strength. Electron-rich regions on the ESP surface coincided with the sites reported as electron-donating in the literature.

github.com/ahadk210-cmyk/Ahad_Khan_Computational_Chemistry

Project 2: Molecular Dynamics

OBJECTIVE

Set up, equilibrate and run an explicit-solvent MD simulation end to end, and verify structural stability of a solute in water over time.

METHODS

GROMACS, GROMOS54A7 force field; 500 ps of ethanol in explicit SPC water (1,252 atoms, 300 K, 1 bar); RMSD and radial distribution function analysis.

RESULTS

System density 975.6 kg/m³; RMSD plateaued at 0.36 Å over the full 500 ps; RDF showed the hydrogen-bonding solvation peak at 2.7 Å.

github.com/ahadk210-cmyk/MD_Project

Project 3: Solvation Effects (CPCM)

OBJECTIVE

Determine whether the gas-phase ordering from Project 1 changes when the same five molecules are placed in an aqueous dielectric environment.

METHODS

Same ORCA/B3LYP/6-31G* setup as Project 1 with the CPCM(water) implicit solvation model; HOMO–LUMO gaps and dipole moments compared against gas phase for all five molecules.

RESULTS

Quercetin and caffeic acid showed reduced gaps in water. Rutin showed the opposite, a 17% gap increase, despite its structural similarity to quercetin; the glycoside substituent is the likely cause, though this was not tested further.

github.com/ahadk210-cmyk/CPCM_Solvation

Input files, scripts, outputs and a README for each project are in the linked repositories.